CUSTOMER CHURN PREDICTION

Final Machine Learning Project Report

Project Area	Machine Learning – Classification
Programming Language	Python
Development Environment	Jupyter Notebook
Dataset	Telco Customer Churn (WA_Fn-UseC_-Telco-Customer-Churn.csv)
Dataset Size	7,043 customer records × 21 columns
Selected Model	Logistic Regression

Prepared from the supplied Customer_Churn_Prediction Jupyter Notebook and its recorded outputs.

1. Executive Summary

This project develops a supervised machine-learning classification workflow to predict whether a telecommunications customer is likely to leave the service (churn). The notebook loads the Telco Customer Churn dataset, performs data inspection and cleaning, explores churn patterns, preprocesses numerical and categorical features, trains two classification models, compares their performance, and demonstrates a prediction for a new customer.

Main result: Logistic Regression achieved 80.55% test accuracy and an F1-score of 0.604 for the churn class. Random Forest achieved 77.00% test accuracy and an F1-score of 0.593. Logistic Regression was therefore selected as the final model based on both accuracy and churn-class F1-score.

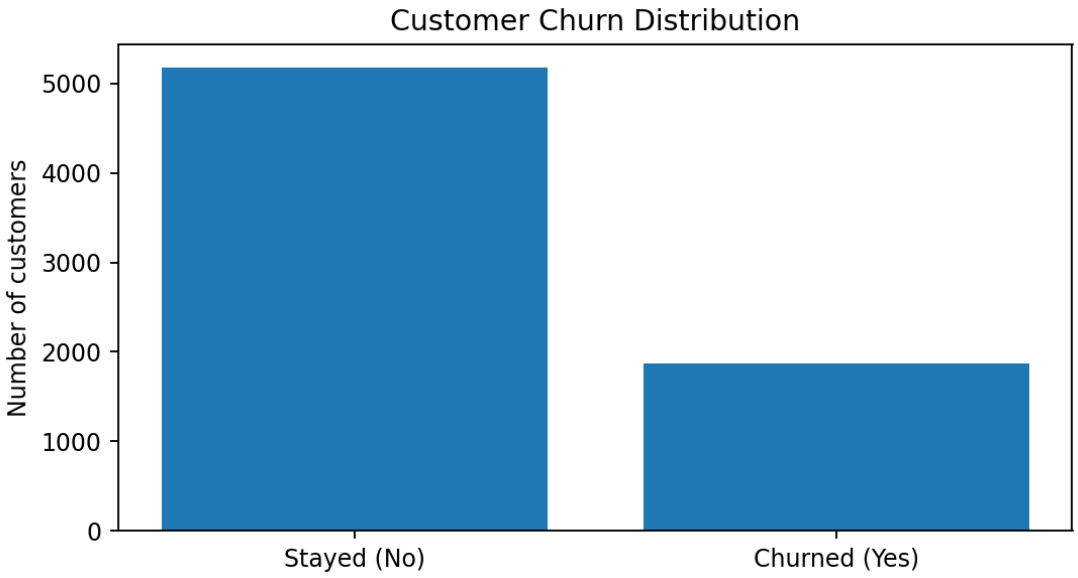
2. Project Objectives

- Understand the customer dataset and its variables.
- Clean missing values and prepare data for machine learning.
- Explore customer churn distribution and selected relationships.
- Train and evaluate Logistic Regression and Random Forest classifiers.
- Select the stronger model and demonstrate a new-customer prediction.
- Save and reload the trained model and preprocessing object.

3. Dataset Description

The notebook uses the Telco Customer Churn CSV file named WA_Fn-UseC_-Telco-Customer-Churn.csv. It contains 7,043 records and 21 columns. The fields cover customer profile, subscribed services, contract and payment information, monthly charges, total charges, and the target variable Churn.

Dataset property	Value
Rows	7,043
Columns	21
Target column	Churn (Yes / No)
Stayed customers	5,174 (73.46%)
Churned customers	1,869 (26.54%)


Customer Churn Distribution
Number of customers
5000
4000
3000
2000
1000
0
Stayed (No)
Churned (Yes)

4. Data Cleaning and Exploratory Data Analysis

Initial inspection found no duplicate rows and no null values before type conversion. The TotalCharges field was initially stored as text. It was converted to numeric using coercion, which exposed 11 missing values. These 11 records had tenure equal to zero; the notebook replaced the missing TotalCharges values with 0. A final verification reported zero missing values and zero duplicate rows, with the dataset retaining 7,043 rows and 21 columns.

Exploratory analysis in the notebook includes:

- Customer churn count distribution.
- Churn by contract type.
- Customer tenure distribution split by churn.
- Monthly charges compared with churn.
- Correlation heatmap for selected numerical fields and encoded churn.

The target distribution is imbalanced: approximately 26.54% of customers churned while 73.46% stayed. This imbalance is important when interpreting accuracy alone.

5. Feature Preparation and Model Training

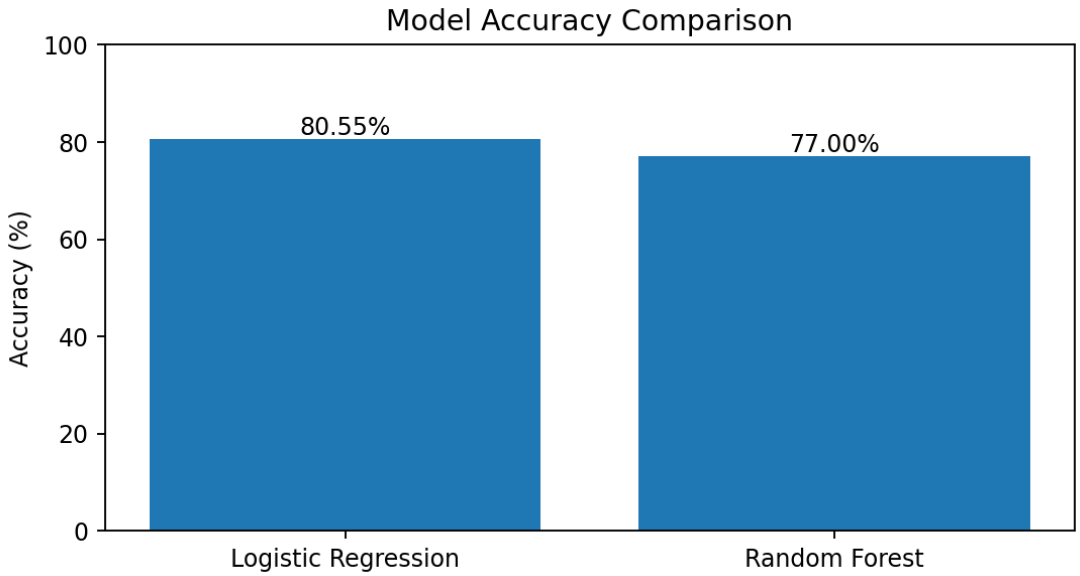
The notebook removes customerID and the target column from the predictors. Churn is mapped to numeric values (No = 0, Yes = 1). The resulting feature matrix contains 19 predictors: 4 numerical features and 15 categorical features.

The data is split into 80% training and 20% testing using train_test_split with random_state=42 and stratify=y. This produced 5,634 training records and 1,409 testing records. Numerical features are standardized with StandardScaler, while categorical features are transformed using OneHotEncoder(handle_unknown='ignore'). The preprocessing object is fitted on training data and then applied to the test data. The transformed matrices contain 45 features.

Two models were trained: Logistic Regression (max_iter=1000, random_state=42) and Random Forest (100 estimators, random_state=42, class_weight='balanced').

6. Model Evaluation and Comparison

Metric	Logistic Regression	Random Forest
Accuracy	80.55%	77.00%
Churn-class precision	0.66	0.56
Churn-class recall	0.56	0.63
Churn-class F1-score	0.604	0.593



Logistic Regression performed better in overall accuracy and churn-class F1-score. Random Forest had higher recall for churn (0.63 compared with 0.56), meaning it identified a larger share of actual churn cases in this test set, but it also had lower precision and overall accuracy. The notebook selects Logistic Regression as the best model using F1-score, which is consistent with the reported values.

7. Sample Customer Prediction

The notebook creates one sample customer with the following profile: Female; not a senior citizen; no partner or dependents; tenure of 2 months; fiber-optic internet; no online security, backup, device protection or tech support; month-to-month contract; paperless billing; electronic-check payment; monthly charges of 70.70; and total charges of 151.65.

Prediction output	Notebook result
Selected model	Logistic Regression
Predicted outcome	Customer May Leave
Estimated churn probability	69.16%

This is a model-generated estimate for the supplied example, not a guarantee that the customer will leave.

8. Model Saving and Loading

The notebook saves a bundle containing the selected model, fitted preprocessor, and model name to customer_churn_project/churn_model.pkl using joblib. It then loads the saved bundle and tests the loaded model on the processed test data. The saved model reproduces the reported test accuracy of 80.55%.

9. Conclusion

The project demonstrates an end-to-end customer churn prediction workflow using Python and scikit-learn. After cleaning and preprocessing the Telco dataset, Logistic Regression achieved the highest reported accuracy (80.55%) and churn-class F1-score (0.604), and was selected as the final model. The sample prediction indicates a 69.16% estimated churn probability for the example customer.

10. Limitations and Future Scope

- The reported evaluation uses a single stratified train/test split; cross-validation could provide a more stable estimate.
- Churn-class recall for the selected Logistic Regression model is 0.56, so some customers who actually churn may not be identified.
- Further work could tune classification thresholds to balance precision and recall according to business priorities.
- Additional models and systematic hyperparameter tuning could be evaluated.
- A deployment interface or dashboard could make predictions easier for business users.
- Model monitoring and periodic retraining could be added when new customer data becomes available.

Appendix: Tools and Libraries

Python; Pandas; NumPy; Matplotlib; Seaborn; scikit-learn; Joblib; Jupyter Notebook. Notebook environment versions recorded: Pandas 3.0.3, NumPy 2.4.4, and scikit-learn 1.9.0.

Report prepared from the supplied notebook outputs. Results and numerical values are reproduced from that notebook; no additional model training was performed for this report.